

Supplementary Material for "TFCDiff: Robust ECG Denoising via Time-Frequency Complementary Diffusion"

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I. SUPPLEMENTARY NOTES

Supplementary Note 1 | Comparison of DCT against FFT and DWT

The Discrete Cosine Transform (DCT) can be viewed as a specialized form of the Discrete Fourier Transform (DFT) that applies even-symmetry extension to the input signal. This preprocessing step confers several critical advantages over the Fast Fourier Transform (FFT) for deep learning applications. Primarily, the real-valued even-symmetry property ensures that the transformation yields purely real coefficients, thereby obviating the need for complex arithmetic. In contrast, FFT generates complex-valued coefficients, which significantly complicates neural network architecture design. Common strategies to handle complex numbers, such as processing real and imaginary components through independent modules or operating solely on magnitudes, invariably increase computational overhead and obscure the modeling of amplitude-phase dependencies. Furthermore, current deep learning frameworks like PyTorch offer limited GPU acceleration for complex operations, posing engineering challenges. Beyond computational efficiency, DCT exhibits superior energy compaction, allowing for more effective signal representation with fewer coefficients. Our preliminary experiments adapting the architecture from [31] for FFT-based inputs confirmed these theoretical drawbacks. The one-generation model incurred prohibitive latency and demonstrated poor denoising performance (SSD = 505.521 μV^2 , MAD = 0.790 mV, PRD = 109.173 %, CosSim = 0.838, ImSNR = 0.786 dB), leading us to discard the FFT approach in favor of DCT.

While the Discrete Wavelet Transform (DWT) offers competitive time-frequency localization through Multi-Resolution Analysis (MRA), it presents specific limitations compared to DCT in the context of ECG denoising. First, the performance of DWT is highly sensitive to the selection of the mother wavelet, making it difficult to identify an optimal basis for ECG signals with heavy noise. Second, the strict dyadic downsampling required by DWT imposes rigid constraints on input signal lengths and reduces the flexibility of coefficient truncation. Specifically, input lengths that are not powers of two often cause compatibility errors within U-Net architectures due to mismatched feature map dimensions. To validate this, we replaced the DCT module with a DWT module using the Daubechies-4 wavelet, applying a 2-level decomposition and retaining only the 0 - 45 Hz approximation coefficients. The resulting one-generation model exhibited inferior denoising quality (SSD = 40.232 $\pm$ 116.930 μV^2 , MAD = 0.521 $\pm$ 0.592 mV, PRD = 39.438 $\pm$ 27.348 %, CosSim = 0.914 $\pm$ 0.110, ImSNR = 11.462 $\pm$ 4.570 dB) and slower inference speeds (82.80 ms vs. 43.72 ms per 10-second segment) compared to the DCT-based baseline. Complementing these statistical findings, we conducted time-frequency domain visualizations via Short-Time Fourier Transform (STFT) spectrograms (Fig. S1) to qualitatively assess signal characteristics. Generated using a Hann window (window length and FFT size of 64 samples; hop length of 16 samples), these spectrograms reveal that while DWT yields a globally smoother background, it inadvertently introduces spurious ECG-like waveforms. Such artifacts constitute a critical deficiency for clinical diagnosis. These empirical results, combined with the structural inflexibility of DWT and the risk of generating diagnostic false positives, substantiate our choice of DCT as the most robust and efficient time-frequency representation for this study.

Supplementary Note 2 | Validation of Signal Fidelity Under Truncation

To verify that truncation does not compromise signal fidelity, we evaluated 37,590 ten-second ECG segments from the QT Database (detailed in Section IV-B). Each segment is transformed via 1D DCT, truncated to retain only coefficients corresponding to frequencies below 50 Hz, zero-padded back to the original length, and reconstructed using 1D IDCT. Quantitative comparison between the original and reconstructed signals yielded an average SSD of $1.68 \times 10^{-3} \mu\text{V}^2$ and an average MAD of 9.01×10^{-3} mV, indicating negligible waveform distortion. Furthermore, the retained low-frequency coefficients accounted for an average of 99.9991 % of the total signal energy. These results conclusively demonstrate that the truncation process preserves essential signal morphology and energy with high precision, validating the signal integrity under truncation.

Supplementary Note 3 | Forward and Reverse Diffusion Process

We are given a pair of DCT coefficient vectors $\{\mathbf{d}_0, \tilde{\mathbf{d}}\}$, where $\mathbf{d}_0$ denotes the clean coefficients and $\tilde{\mathbf{d}}$ represents their noisy counterparts, both drawn from an unknown conditional data distribution. Our goal is to learn a model that approximates this conditional distribution, thereby mapping a noisy input $\mathbf{d}_{noisy}$ to its corresponding noise-free output $\mathbf{d}_{clean}$. To this end, we formulate a discrete time conditional diffusion model.

For a discrete timestep $t \in \{1, 2, \dots, T\}$, the *forward diffusion process* q perturbs the clean coefficients $\mathbf{d}_0$ according to:

$$\mathbf{d}_t = \sqrt{\bar{\alpha}_t} \mathbf{d}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad (1)$$

where $\mathbf{d}_t$ is the latent representation at step t , $\bar{\alpha}_t \in (0, 1]$ is determined by a predefined noise schedule (detailed in Section III-C), and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$.

When $t = T$, $\mathbf{d}_T$ approximates isotropic Gaussian noise. The forward perturbation kernel $q(\mathbf{d}_t | \mathbf{d}_0)$ is Gaussian:

$$q(\mathbf{d}_t | \mathbf{d}_0) = \mathcal{N}(\mathbf{d}_t; \sqrt{\bar{\alpha}_t} \mathbf{d}_0, \sqrt{1 - \bar{\alpha}_t} \mathbf{I}). \quad (2)$$

In the *reverse process* p , we iteratively recover the clean coefficients by denoising $\mathbf{d}_T$ using a parameterized model $p(\mathbf{d}_{t-1} | \mathbf{d}_t, \tilde{\mathbf{d}})$, where the observed noisy coefficients $\tilde{\mathbf{d}}$ serves as a conditioning input. This reverse process is also modeled as a Gaussian:

$$p(\mathbf{d}_{t-1} | \mathbf{d}_t, \tilde{\mathbf{d}}) = \mathcal{N}(\mathbf{d}_{t-1}; \mu_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}), \sigma_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}) \cdot \mathbf{I}), \quad (3)$$

with mean and standard deviation defined as:

$$\mu_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}) = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(\mathbf{d}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}) \right), \quad (4)$$

$$\sigma_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}}) = \begin{cases} \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t, & \text{if } t \geq 1 \\ \beta_1, & \text{if } t = 1 \end{cases}, \quad (5)$$

where $\epsilon_\theta(\mathbf{d}_t, t, \tilde{\mathbf{d}})$ is a U-Net-based noise predictor that predicts the noise component ϵ .

Importantly, all inputs in this framework are truncated DCT coefficients as described in Section III-A. After denoising, the output coefficients are zero-padded to their original length and transformed back to the time domain via the 1D IDCT to yield the final denoised ECG signal.

Supplementary Note 4 | Definitions of Evaluation Metrics

SSD, MAD, PRD, CosSim and ImSNR are defined as follows, with h_0 denoting the ground truth signal, $\tilde{\mathbf{l}}$ and $\hat{\mathbf{l}}$ representing the noisy and denoised signals respectively, and N indicating the signal length.

$$\text{SSD}(\mathbf{l}_0, \hat{\mathbf{l}}) = \sum_{i=1}^N [\mathbf{l}_0(i) - \hat{\mathbf{l}}(i)]^2, \quad (6)$$

$$\text{MAD}(\mathbf{l}_0, \hat{\mathbf{l}}) = \max |\mathbf{l}_0(i) - \hat{\mathbf{l}}(i)|, \quad i \in \{1, 2, \dots, N\}, \quad (7)$$

$$\text{PRD}(\mathbf{l}_0, \hat{\mathbf{l}}) = \sqrt{\frac{\sum_{i=1}^N [\mathbf{l}_0(i) - \hat{\mathbf{l}}(i)]^2}{\sum_{i=1}^N [\hat{\mathbf{l}}(i) - \frac{1}{N} \sum_{i=1}^N \mathbf{l}_0(i)]^2}}, \quad (8)$$

$$\text{CosSim}(\mathbf{l}_0, \hat{\mathbf{l}}) = \frac{\langle \mathbf{l}_0, \hat{\mathbf{l}} \rangle}{\|\mathbf{l}_0\|_2 \|\hat{\mathbf{l}}\|_2}, \quad (9)$$

$$\begin{aligned} \text{ImSNR}(\hat{\mathbf{l}}, \tilde{\mathbf{l}}) &= \text{SNR}(\mathbf{l}_0, \hat{\mathbf{l}}) - \text{SNR}(\mathbf{l}_0, \tilde{\mathbf{l}}) \\ &= 10 \log_{10} \left(\frac{\sum_{i=1}^N [\mathbf{l}_0(i) - \tilde{\mathbf{l}}(i)]^2}{\sum_{i=1}^N [\mathbf{l}_0(i) - \hat{\mathbf{l}}(i)]^2} \right). \end{aligned} \quad (10)$$

Supplementary Note 5 | Definitions of Fiducial Point Detection Metrics

The performance indicators of fiducial point detection are F1-score (F1) and Mean Absolute Error (MAE), which are defined as follows. Let $P_{\text{gt}} = \{p_1, p_2, \dots, p_M\}$ denote the set of ground truth fiducial point positions detected in the clean signal, and $P_{\text{pred}} = \{\hat{p}_1, \hat{p}_2, \dots, \hat{p}_K\}$ denote the set of predicted fiducial point positions in the denoised signal.

To account for physiological tolerance and minor temporal shifts, a predicted fiducial point $\hat{p}_k$ is considered a True Positive (TP) if it falls within a fixed tolerance window of $\delta = 200$ ms around an unmatched ground truth fiducial point p_m . Formally, a match exists if:

$$|\hat{p}_k - p_m| \leq \delta, \quad (11)$$

where each ground truth fiducial point is matched to at most one prediction. Predictions without a corresponding ground truth fiducial point within δ are counted as False Positives (FP), while ground truth fiducial points without a matching prediction are counted as False Negatives (FN).

Based on these counts, F1-score for a specific wave type is calculated as the harmonic mean of Precision and Recall:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (12)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (13)$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (14)$$

MAE quantifies the temporal precision of the detected fiducial points while penalizing missed detections to reflect clinical utility. Unlike standard MAE calculations that only average errors over matched pairs, our metric incorporates a penalty for FN based on the average cardiac cycle length. This penalty mechanism prevents models that tend to predict fiducial points only on simple samples from achieving artificially lower MAE scores. Specifically, it is computed as:

$$\text{MAE} = \frac{1}{N_{\text{gt}}} \left(\sum_{i \in \mathcal{M}} |p_i^{(\text{gt})} - \hat{p}_i^{(\text{pred})}| + \text{FN} \cdot \overline{RR} \right), \quad (15)$$

where N_{gt} denotes the total number of ground truth fiducial points, $\mathcal{M}$ denotes the set of indices for successfully matched pairs, $p_i^{(\text{gt})}$ and $\hat{p}_i^{(\text{pred})}$ represent the time indices of the i -th matched ground truth and predicted points, respectively, and $\overline{RR}$ is the average RR interval of the signal. A lower MAE value indicates both high detection completeness and precise temporal alignment.

In our experiments, both metrics are computed separately for P, Q, R, S, and T waves, and the final reported values represent the macro-average across all test samples in the synthesized dataset.

II. SUPPLEMENTARY FIGURES

figures/DWT.pdf

Fig. S1. Time-frequency domain analysis via STFT spectrograms comparing the denoising performance of applying DCT and DWT. (a) Clean ground truth. (b) Noisy input. (c) The signal denoised by TFCDiff with DCT. (d) The signal denoised by TFCDiff with DWT.

figures/coefficients.pdf

Fig. S2. Distributions of the DC and AC components. Maroon lines indicate the 1st, 5th, 10th, 50th, 90th, 95th, and 99th percentiles for both components.

figures/Synthesized_spectrogram.pdf

Fig. S3. Time-frequency domain analysis of TFCDiff denoised results on the synthesized dataset via STFT spectrograms generated using a Hann window (window length and FFT size of 64 samples, hop length of 16 samples). (a) Noise Level: 0.2-0.6. (b) Noise Level: 0.6–1.0. (c) Noise Level: 1.0–1.5. (d) Noise Level: 1.5–2.0.

figures/SimEMGdisplay.pdf

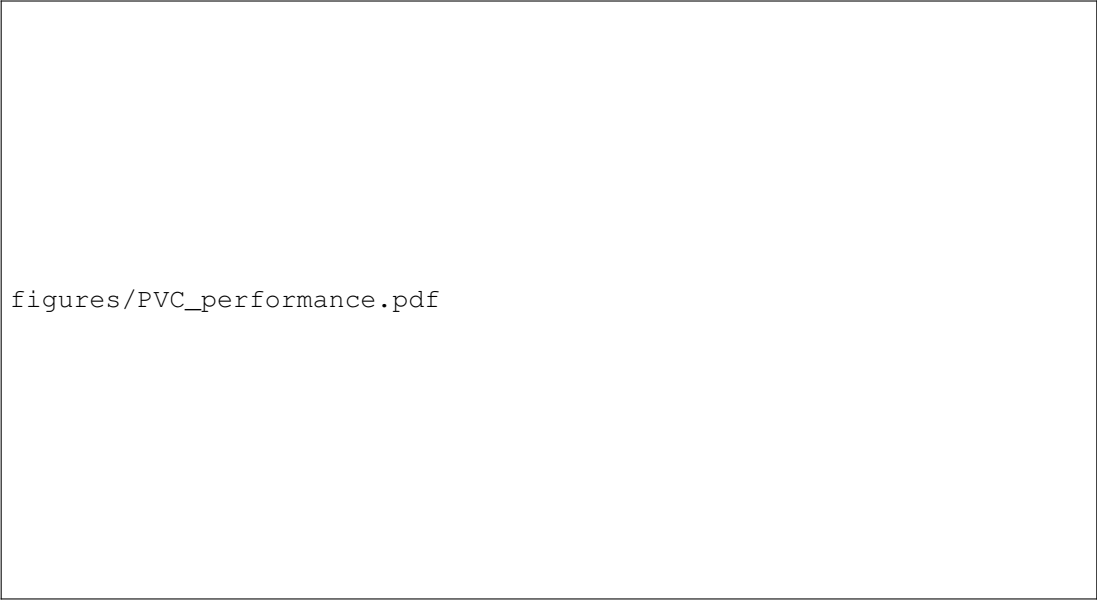
Fig. S4. Visualization of TFCDiff denoised results on the SimEMG Database with representative samples. (a) Input SNR 2.560 dB: SSD 2.644 μV^2 , MAD 0.175 mV, PRD 19.258 %, CosSim 0.987, ImSNR 10.681 dB. (b) Input SNR -0.130 dB: SSD 1.684 μV^2 , MAD 0.137 mV, PRD 22.995 %, CosSim 0.975, ImSNR 12.208 dB.

figures/simemg_spectrogram.pdf

Fig. S5. Time-frequency domain analysis of TFCDiff denoised results on the SimEMG Database via STFT spectrograms generated using a Hann window (window length and FFT size of 64 samples, hop length of 16 samples). (a) Input SNR: 2.560 dB. (b) Input SNR: -0.130 dB.

figures/ablation_spectrogram.pdf

Fig. S6. Time-frequency domain analysis of denoised results in the ablation study of DCT diffusion and TFEM via STFT spectrograms generated using a Hann window (window length and FFT size of 64 samples, hop length of 16 samples). (a) Ground truth signal. (b) Noisy ECG signal. (c) Time-domain diffusion only. (d) Time-domain diffusion with TFEM. (e) DCT diffusion only. (f) DCT diffusion with TFEM.



figures/PVC_performance.pdf

Fig. S7. Visualization of TFCDiff denoised results on PVC signals across varying noise intensities.



figures/prototype.pdf

Fig. S8. Prototype of the developed wearable ECG monitoring system, showcasing the host software interface for real-time data acquisition and synchronized denoising.

III. SUPPLEMENTARY TABLES

TABLE S1
SUMMARY OF DATASETS

Dataset	QTDB	NSTDB	SimEMG
Recordings	105	3	220
Length (second)	900	1800	30
Channel	2	2	1
Fs (Hz)	250	360	500

TABLE S2
RECORDINGS SELECTED FOR TEST SET

Database	Recording ID
MIT-BIH Arrhythmia Database	sel123, sel233
MIT-BIH ST Change Database	sel302, sel307
MIT-BIH Supraventricular Arrhythmia Database	sel820, sel853
MIT-BIH Normal Sinus Rhythm Database	sel16420, sel16795
European ST-T Database	sel0106, sel0121
Sudden death patients from BIH	sel32, sel49
MIT-BIH LongTerm ECG Database	sel14046, sel15814

TABLE S3
SSD (μV^2) COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

Models	0.2-0.6	0.6-1.0	1.0-1.5	1.5-2.0
FIR	17.414 $\pm$ 38.569	53.222 $\pm$ 99.580	125.252 $\pm$ 260.503	233.005 $\pm$ 465.136
IIR	23.732 $\pm$ 62.589	58.063 $\pm$ 108.847	128.126 $\pm$ 259.979	234.929 $\pm$ 463.763
DRNN	42.077 $\pm$ 97.849	55.626 $\pm$ 125.557	75.556 $\pm$ 171.716	98.773 $\pm$ 208.981
FCN-DAE	28.308 $\pm$ 49.248	32.132 $\pm$ 54.977	39.145 $\pm$ 72.682	50.431 $\pm$ 86.602
Multibranch LANLD	21.255 $\pm$ 36.343	29.227 $\pm$ 47.199	39.379 $\pm$ 69.582	53.981 $\pm$ 90.000
DesCod-1	16.599 $\pm$ 40.581	24.596 $\pm$ 61.984	36.821 $\pm$ 93.622	52.919 $\pm$ 116.625
DesCod-3	12.253 $\pm$ 39.038	18.544 $\pm$ 57.313	27.147 $\pm$ 73.191	38.675 $\pm$ 86.161
DesCod-5	11.114 $\pm$ 34.247	17.600 $\pm$ 53.325	25.457 $\pm$ 71.325	36.858 $\pm$ 88.830
DesCod-10	10.708 $\pm$ 36.446	16.266 $\pm$ 51.675	23.838 $\pm$ 66.124	35.127 $\pm$ 84.742
TCDAE	10.703 $\pm$ 23.814	15.012 $\pm$ 32.874	22.207 $\pm$ 52.657	33.138 $\pm$ 72.822
TFCDiff-1	13.215 $\pm$ 49.785	20.622 $\pm$ 63.198	30.499 $\pm$ 89.642	42.378 $\pm$ 107.589
TFCDiff-3	10.791 $\pm$ 44.262	15.421 $\pm$ 48.989	23.594 $\pm$ 76.668	33.245 $\pm$ 89.130
TFCDiff-5	10.153 $\pm$ 42.118	14.676 $\pm$ 48.814	22.430 $\pm$ 72.247	31.061 $\pm$ 83.895
TFCDiff-10	9.677$\pm$40.306	14.222$\pm$46.310	21.512$\pm$70.509	29.497$\pm$79.668

TABLE S4
MAD (mV) COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

Models	0.2-0.6	0.6-1.0	1.0-1.5	1.5-2.0
FIR	0.242±0.223	0.440±0.346	0.665±0.536	0.947±0.763
IIR	0.332±0.439	0.499±0.417	0.701±0.580	0.971±0.778
DRNN	0.425±0.466	0.517±0.513	0.641±0.585	0.775±0.599
FCN-DAE	0.481±0.473	0.488±0.428	0.542±0.488	0.603±0.464
Multibranch LANLD	0.280±0.293	0.347±0.294	0.411±0.330	0.499±0.368
DesCod-1	1.056±1.095	1.045±1.071	1.095±1.054	1.212±1.075
DesCod-3	0.686±0.589	0.719±0.610	0.763±0.597	0.834±0.606
DesCod-5	0.580±0.481	0.611±0.488	0.650±0.489	0.727±0.521
DesCod-10	0.458±0.394	0.490±0.403	0.517±0.416	0.618±0.474
TCDAE	0.216±0.266	0.258±0.253	0.337±0.334	0.448±0.399
TFCDiff-1	0.214±0.377	0.287±0.360	0.376±0.434	0.490±0.474
TFCDiff-3	0.193±0.368	0.255±0.320	0.332±0.396	0.429±0.437
TFCDiff-5	0.190±0.361	0.249±0.318	0.323±0.387	0.416±0.417
TFCDiff-10	0.186±0.347	0.244±0.319	0.316±0.381	0.404±0.406

TABLE S5
PRD (%) COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

Models	0.2-0.6	0.6-1.0	1.0-1.5	1.5-2.0
FIR	26.222±8.911	42.957±10.788	57.398±12.384	68.943±11.535
IIR	29.103±10.469	44.839±10.996	58.331±12.181	69.591±11.374
DRNN	54.682±26.012	63.227±26.219	72.672±25.243	82.869±22.559
FCN-DAE	44.133±32.204	45.370±25.707	52.251±29.742	58.609±25.216
Multibranch LANLD	37.009±16.882	45.729±18.433	55.301±21.659	63.835±19.963
DesCod-1	27.457±14.801	33.635±16.180	42.735±21.211	52.751±23.157
DesCod-3	22.428±11.772	28.613±14.647	37.266±20.029	46.841±22.564
DesCod-5	20.896±10.368	27.414±14.705	35.719±19.796	45.277±23.174
DesCod-10	19.553±10.260	25.796±13.824	34.238±19.345	44.142±23.278
TCDAE	21.258±9.321	25.393±9.537	31.389±12.939	38.027±15.648
TFCDiff-1	19.003±11.936	26.624±15.408	34.451±20.583	43.463±25.293
TFCDiff-3	16.310±11.206	22.798±13.401	30.054±19.061	38.986±25.401
TFCDiff-5	15.645±11.059	21.951±12.547	29.166±19.576	37.790±24.721
TFCDiff-10	15.179±10.830	21.358±12.943	28.390±19.596	36.726±24.850

TABLE S6
COSSIM COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

Models	0.2-0.6	0.6-1.0	1.0-1.5	1.5-2.0
FIR	0.961±0.028	0.895±0.056	0.805±0.091	0.708±0.108
IIR	0.951±0.041	0.886±0.059	0.799±0.091	0.702±0.109
DRNN	0.922±0.037	0.884±0.057	0.825±0.087	0.749±0.111
FCN-DAE	0.919±0.090	0.913±0.076	0.886±0.098	0.853±0.104
Multibranch LANLD	0.950±0.026	0.924±0.038	0.887±0.063	0.843±0.079
DesCod-1	0.953±0.067	0.938±0.068	0.903±0.095	0.860±0.115
DesCod-3	0.972±0.035	0.958±0.043	0.933±0.066	0.901±0.086
DesCod-5	0.977±0.024	0.963±0.037	0.941±0.056	0.911±0.079
DesCod-10	0.980±0.022	0.968±0.031	0.947±0.052	0.918±0.072
TCDAE	0.977±0.024	0.967±0.028	0.949±0.047	0.924±0.068
TFCDiff-1	0.980±0.030	0.963±0.044	0.939±0.069	0.906±0.102
TFCDiff-3	0.985±0.025	0.974±0.029	0.956±0.052	0.930±0.080
TFCDiff-5	0.986±0.023	0.976±0.026	0.959±0.050	0.937±0.071
TFCDiff-10	0.987±0.022	0.977±0.025	0.962±0.048	0.941±0.067

TABLE S7
IMSNR (dB) COMPARISON RESULTS IN DIFFERENT NOISE LEVELS

Models	0.2-0.6	0.6-1.0	1.0-1.5	1.5-2.0
FIR	5.265±3.260	6.388±3.281	6.825±3.502	6.890±3.400
IIR	4.375±3.726	5.987±3.296	6.676±3.519	6.816±3.397
DRNN	1.441±3.925	6.387±3.176	8.898±2.876	10.316±2.553
FCN-DAE	2.388±4.042	8.110±3.311	11.062±3.125	12.758±2.921
Multibranch LANLD	3.597±3.906	8.156±3.315	10.671±3.216	12.191±2.842
DesCod-1	5.630±5.092	10.234±4.164	12.299±4.019	13.307±3.716
DesCod-3	7.464±4.521	11.832±3.938	13.888±3.905	14.907±3.686
DesCod-5	8.067±4.339	12.289±3.883	14.376±3.808	15.363±3.678
DesCod-10	8.690±4.230	12.846±3.811	14.852±3.793	15.722±3.721
TCDAE	7.706±4.385	12.127±3.618	14.300±3.466	15.549±3.301
TFCDiff-1	8.995±3.881	12.410±3.514	14.333±3.478	15.287±3.516
TFCDiff-3	10.504±3.901	13.922±3.679	15.779±3.547	16.716±3.670
TFCDiff-5	10.905±3.939	14.276±3.695	16.139±3.597	17.094±3.704
TFCDiff-10	11.217±3.983	14.601±3.741	16.454±3.658	17.412±3.765